

Given an acute triangle ABC with $\angle CBA = 75^\circ$, orthocenter H , and circumcircle Ω , we analyze the problem involving points H_A , H_C , X , P , and Q . The goal is to determine the values x and z related to the growth rate of the sequence of amazing odd integers $n_1 < n_2 < \dots$, and express $x + 2016z$ in the form $p + q\sqrt{r}$.

Key steps and reasoning:

1. **Triangle Setup and Coordinates:** The triangle is set with B at the origin, BC along the x-axis, and coordinates for A , H , and other points derived using trigonometric identities and properties of the orthocenter.
2. **Circle and Intersections:** The circle centered at B with radius BH intersects BC at X . Lines XH_A and CH intersect at P , while lines XH_C and AH intersect at Q .
3. **Ratio Condition:** The ratio $\frac{CP}{CH} = \sqrt{n}$ must be satisfied, leading to complex coordinate and trigonometric calculations.
4. **Integer Lengths:** For n to be amazing, AQ and H_CQ must be integers, implying solutions to Diophantine equations.
5. **Growth Rate Analysis:** The sequence n_i grows exponentially. The problem requires finding the largest real number x such that $n_{2i} \geq yx^i$ and the largest z such that $n_{2i} \geq zx^i$.
6. **Pell Equation Insight:** The growth rate is linked to solutions of a Pell equation, leading to the dominant root involving $\sqrt{3}$. The characteristic equation's roots suggest the growth rate $x = (2 + \sqrt{3})^2 = 7 + 4\sqrt{3}$.
7. **Final Calculation:** Assuming $z = 1$ (the constant term in the asymptotic growth), we compute $x + 2016z = 7 + 4\sqrt{3} + 2016$, resulting in $2023 + 4\sqrt{3}$. Thus, $p = 2023$, $q = 4$, and $r = 3$, giving $p + q + r = 2030$.

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